Scale-free dynamics of visual inference

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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

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Abstract

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Acknowledgements

Contents

Introduction

The brain is among the most widely studied subjects of research, and yet many neural behaviours remain elusive to scientists. In recent years, methods adapted from statistical physics, network science, and thermodynamics have been found to explain and provide understanding for computation in the brain. A large part of the brain is devoted to vision, the visual cortex containing between 4-6 billion of the 30 billion neurons of the neocortex, up to 5 times the size of other primates [1, 2]. Vision is thus an important neural process, and provides one of the first gates of attention to the human brain.

Many approaches to understanding patterns of human gaze have been described. Importantly, there has been success in training large neural networks to predict gaze trajectories [3–8] and saliency maps from an image for human observers [9–14]. Such approaches give often quite accurate results, but they lack the rich understanding of a full model of the network of the brain itself. Theories of biological attention and cognition, such as those of the "Bayesian brain", built on active sensing and inference provide a more motivated framework to understand the brain as maintaining a rational causal mental worldview [14–17]. This Bayesian model of the brain has been a fruitful direction to understand the most basic level of neural cognition as sampling through population activity patterns [18–21]. Although it has been a debated framework (see [22]), the Bayesian brain has provided a strong window into biological cognition. Geoffrey Hinton and colleagues invented the Helmholtz machine, performing explicit Bayesian inference on perceptive inputs, and updating predictions on the external world, minimising the free energy as a loss function of a neural network. Such a formulation inspired pioneering work on neural coding [20, 23], Karl Friston's highly influential Free Energy Principle in theoretical neuroscience [24, 25], and the resulting empirical models and well-verified predictions about the human sensory-motor system [26–28].

Chapter 1: Survey of the Literature

This thesis aims to further our understanding of the details of biological neural computation through understanding the dynamics of visual searching. There is growing success in turning methods from statistical physics towards the area of computational neuroscience, and major contributions to this project are described here. In particular, the rich scale-free activity of neural circuits inspires similar analysis to the dynamics of gaze trajectories, alongside the information-theoretic pressure of the visual system to process data near-optimally. We aim to understand these behaviours as an efficient method of gathering information. In Section 1.1 we introduce the framework of stochastic modelling, Lévy motion, and the associated anomalous superdiffusion which informs the dynamics of the FNS sampler. Section 1.2 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts. Section 1.3 provides the background for much of the behavioural data we present and model in this thesis, and the previous methods used to understand and model the complex dynamics found in gaze trajectories especially.

1.1 Stochastic Modelling

1.1.1 Stochastic Calculus

Stochastic calculus has been fundamental in understanding how high dimensional complex systems reduce to a small number of variables of interest. When studying complex systems in physics, economics, geology, climate science, machine learning, and neuroscience, it is often the most natural, tractable, and accurate modelling choice when understanding a system of interest. The important concept here is that of the introduction of some central limit theorem, where the sum of very many small interactions which are understood microscopically can produce certain well-defined stochastic patterns on macroscopic scales of interest. The ubiquitous stochastic process is a Wiener process W_t, equivalently called Brownian Motion. A Wiener process is defined by the stationary and independent Gaussian increments across all non-overlapping time intervals. The Wiener process is a very simple and analytically elegant model of the coarse-grained effect of very many small finite-variance interacting influences. A Wiener process is defined only by its timescale, with a variance over trajectories after time t of t itself. Wiener processes yield time series which are almost surely continuous but nowhere differentiable. The Wiener process is understood as a model of the long-time dynamics of a variable effected by very many small uncorrelated finite variance influences [29]. Classical examples of processes modelled by the Wiener process are: the movement of a macroscopic particle in a non-turbulent fluid [29], the gravitational influence of an orbiting thermal collection of stars on supermassive black holes [30, 31], and (arguably) the movement of low-volatility stock prices [32]. There are well studied mechanisms through which complex real-world stochastic processes diverge from the Wiener process. Of interest to us is the case where the microscopic effects become correlated with heavy-tailed statistics. Correlated steps, long-time memory effects, and microscopic dynamics can result in a stochastic process which does not limit to the Wiener process [33, 34]. Of interest to us is the fractional and superdiffusive Lévy flight, which is a result of spatial non-locality in an interacting network. For instance, previous studies have demonstrated a neural circuit network at criticality resulting in diverging correlation length of activity, and the centre of mass of neural firing is subject to the large jumps. Such large jumps in a stochastic time series are modelled according to

Lévy stable motion, where the increments are drawn independently from a heavy-tailed distribution. This section describes the analytics and behaviour of such Lévy processes for our current study.

1.1.2 Anomalous Superdiffusion

A key motivation for the development for stochastic modelling was when studying processes of diffusion. In 1885, the German physiologist Adolf Fick was the first to describe the universality of normal diffusion [35]. Fick records the diffusion equation, which is identical to the heat equation applied to a probability distribution, as

$$\frac{\partial p(x,t)}{\partial t} = D\frac{\partial^2 p(x,t)}{\partial x^2} \tag{1.1}$$

where D is a scaling constant. Fick shows that this gives a mean squared displacement (MSD) $\langle x(t)^2 \rangle = 2Dt$. This MSD linear in time defines normal diffusion. Such normal diffusion is found in many simple random processes, such as the paradigm case of the Brownian motion of dust in a fluid studied by Albert Einstein. Both Albert Einstein and the British statistician Karl Pearson independently developed the continuous time random walk (CTRW) with finite-variance step size and step time distributions, robustly producing this normal diffusion. Increasingly, physical systems have been which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle x(t)^2 \rangle \propto t^\beta, \quad \beta \neq 1 \tag{1.2}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \beta < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \beta < 2$ where diffusion scales faster than normal. The case of $\beta = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$. Anomalous subdiffusion has been shown to result from heavy-tailed wait-times between interactions [35], such as in trapped fluid flow [36], controlled diffusion through cell membranes [35], and protein mixing [37].

Anomalous superdiffusion

More importantly to this thesis, superdiffusion has been studied in the contexts of complex exploratory biological systems. Superdiffusion has a strong link to the problem of a blind search for scarce resources. The Lévy foraging hypothesis posits that a superdiffusive CTRW (known as a Lévy walk) is an optimal strategy when searching for sparse and randomly distributed resources. This result was first reported in a letter to the journal Nature in 1999 by Viswanathan et al. [38]. In this paper a Lévy walk with superdiffusion $\langle x(t)^2 \rangle \propto \frac{t^2}{\ln(t)}$ on the transition to ballistic diffusion was found to be optimal in a search for sparse resources. The optimality of the superdiffusive Lévy walk has been reproduced in diverse functional search models [39–41], in a population of searching agent [42], and in the empirical search paths of biological organisms, from the flights of drosophilia flies [41, 43], albatross' searches [44, 45], and strongly across many marine predators [46, 47]. Increasingly there are studies which describe biological limitations of the Lévy foraging hypothesis, however the Lévy computational advantage in a diverse set of resource landscapes without the need for fine-tuning remains strong [39, 40]. This marks a split in the treatment of Lévy walks, with groups studying them as informative baseline functional models of abstract search [38, 39, 42, 48], and others led by authors such as Graham Pyke presenting this as a misrepresentation of true intelligent organisms [49–51], see Reynolds

(2015) for an in-depth discussion of a richer view of biological Lévy walk superdiffusion [52].

1.1.3 Lévy Stable Motion

A general process which is highly analytically tractable, not least because it is Markovian, is the *Lévy flight*. The nomenclature in the literature is often unclear when distinguishing stochastic processes of a heavy-tailed nature. Here, the Lévy flight L_T^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution). $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables,as a result of the generalised central limit theorem. The α value defining a particular $\mathcal{S}\alpha\mathcal{S}$ distribution is a result of the asymptotic power-law tail of the density function, with $p(x) \propto |x|^{-1-\alpha}$. Thus, the Lévy flight L_t^α after time t is defined as follows, to outline the fractional scaling properties and alpha-stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.3}$$

where the parameter α defines the heaviness of the tails of resulting statistics. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaling of Brownian Motion $L_t^{\alpha\to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges, but L_t^α has 0 mean, whereas if $0 < \alpha \leq 1$, then both the variance and mean of L_t^α diverge. Many of the simple analytic properties of the Wiener process are no longer true of a Lévy flight. Importantly, trajectories of a Lévy flight are almost surely no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite á gauche) with respect to time, which describes functions which are right continuous with well-defined left limits [53]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [54, 55]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [53]. Importantly, it follows from the theorem of the Lévy-Itô decomposition that a Lévy flight can be decomposed as the sum of two processes [56]:

1. A compound Poisson process of large jumps with $|\Delta x| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number of such jumps in any finite time period.

2. A square-integrable martingale jump process with $|\Delta x| < 1$. This process is thus finite-variance, and when coarse-grained over time, approaches a Wiener process.

This division is useful in analysing the process in the context of stochastic nonlocal global exploration, which is modelled by the compound Poisson process of large jumps $|\Delta x| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta x| < 1$.

1.2 Abstract Computation

Probabilistic sampling is a powerful and flexible paradigm of general computation. Sampling methods represent and evaluate probability distributions through the production of stochastic samples samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods draw from a target distribution such that the sampled ensemble distribution reflects the target itself. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from large multivariate distributions inferred from observational data [57, 58]. This framework has gained significant traction as a basic scheme to describe biological computation. The position that the brain itself acts as a Bayesian generative model has become that interprets through continuous sampling mechanisms has become well supported.

1.2.1 Active Searching and Inference

The dominant framework in modern computational neuroscience is that of the Bayesian brain. Karl Friston, the most cited neuroscientist [59], developed this framework, and shifted the view of the perceptual loop of the brain. Friston's theory of active inference described perception through inferring and generating top-down sensory predictions of the world, and carrying back the bottom-up errors from that perception. This development brought a strong shift in the literature, "marking a watershed between 20th-century thinking about the brain as a glorious stimulus–response link and more constructivist 21st century perspectives that emphasized an active sampling of the sensory world" ([60] pg. 1019). The active inference model of an updating internal mental state filled the explanatory gap in the understanding of the functional role of many key feedback mechanisms in the brain [27, 28], and went on to yield successful predictions of how the brain optimally deciphers noisy multimodal sensory inputs [26], as well as provide rigorous frameworks to base probabilistic computation in the brain [61].

There has recently been opposition to the general framework of the Bayesian brain and active inference [22]. Much of the

1.2.2 Neural Sampling

The neural sampling hypothesis posits that the variability of a cortical neural response is an important functional feature of the biological computational sampling. Previously regarded as noise, the neural sampling framework interprets instantaneous neural activity patterns as stochastic samples drawn from a posterior distribution over possible sensory interpretations [18, 19]. As such, the location and frequency of firing patterns across cortical neurons are interpreted as stochastically sampling an informed target posterior, acting as the basis for some kind of computation. Consequently, perceptual uncertainty is directly encoded by the variability of cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [18, 19, 57]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of computation. Various biological mechanisms have been proposed to explain how neural circuits generate samples.

Traditional models often assume the existence of explicit stochastic noise sources, however recent models demonstrate that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models marginalise over through chaotic dynamics. These emergent chaotic dynamics facilitate Bayesian learning, as for instance described by Terada and Toyoizumi through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [62]. Furthermore, energy-based models (EBMs) have been developed which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [63]. The sampling hypothesis has thus given a convincing theory of biological neural computation when contrasted with the trial-averaged perspective of some averaged response combined with noise linked to neural variability [18]. However, traditional Gaussian or Poisson noise models representing these stochastic network dynamics inherently correspond to continuous Brownian motion and struggle to capture the full dynamics of a complex interconnected inhibitory-excitatory biological neuronal network [64–66].

Fractional Neural Sampling

In reality, neural activity exhibits structure across multiple spatial and temporal scales and exhibit heavy-tailed, non-Gaussian statistics. This observation motivates the design and theory of the theory of Fractional Neural Sampling (FNS). FNS describes the complex spatiotemporal dynamics of spiking neural circuits, where localised activity patterns propagate not through ordinary diffusion, but via superdiffusive Lévy flights. Such dynamics are called fractional, as they are the result of nonlocal interactions requiring fractional calculus to describe [67]. These which can be mathematically characterised by stochastic differential equations with fractional order derivatives. The critical advantage of FNS over standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—a major challenge when processing natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [68–71]. This dynamical switching provides a mechanistic explanation for the bimodality of estimate distributions observed in human perceptual switching and visual perception inference.

1.2.3 Markov Chain Monte Carlo Methods

Markov Chain Monte Carlo (MCMC) methods constitute the algorithmic foundation for generating samples from target distributions in Bayesian inference and machine learning. MCMC operates by constructing a Markov chain whose equilibrium distribution is the desired posterior distribution. In the context of neural computation, biological circuits are often modelled as implementing continuous-time variants of MCMC, most notably Langevin sampling and Hamiltonian Monte Carlo (HMC). Langevin dynamics utilise local gradient information from the probability landscape, combined with Gaussian white noise, to explore the state space over time. Despite their ubiquity, traditional MCMC methods encounter severe theoretical and practical limitations when dealing with multimodal distributions. Standard Langevin and HMC dynamics are driven by continuous Brownian and Hamiltonian motion respectively, and the traversal of low-probability regions, equivalent to dynamical energy barriers, becomes exponentially slow. Consequently, the mean exit time for the sampler to leave one mode and discover another grows exponentially with the modal separation distance. To accelerate this sluggish exploration, algorithms often introduce auxiliary momentum variables. However, even with momentum-based acceleration, traversing deep energy valleys remains highly inefficient [68, 69]. The fractional dynamics introduced in FNS-style sampling are one area of research into how fast mixing of samples in complex distributions can be achieved without the need for non-trivial parameter tuning. For instance, algorithms based on Hamiltonian Monte Carlo sampling and second-order Langevin sampling very similar to FNS dynamics have been briefly developed and explored [70, 72, 73]. Under FNS dynamics, the mean exit time between modes scales linearly rather than exponentially with separation, demonstrating a profound computational advantage for efficient biological Bayesian inference [73].

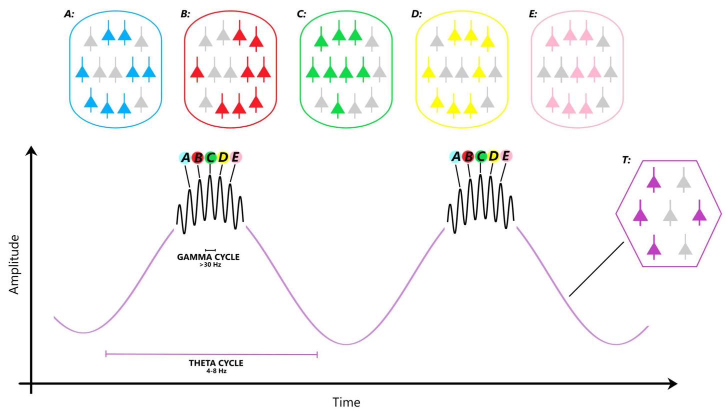

Figure 1.1: A schematic representation of theta-gamma coupling (TGC). TGC has it's foundations in the work of Lisman and Jensen in correctly ordering neural signals [86], and subsequently expanded and used to describe the dynamic mechanism the brain uses during exploration to modulate close attention upon individual centres of focus [87]. *Schematic adapted from figure 2 of M. Ursino and G. Pirazzini, 2024 [83]*

1.3 Behavioural Data

[1/f POWER SPECTRUM]

1.3.1 Neural Frequency Bands

Throughout much of this thesis, links are drawn between timescales of interactions and dynamics in the FNS model and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data. Divisions are only approximate, but for this thesis these bands are: delta (0.1 - 4 Hz), theta (4 - 8 Hz), alpha (8 - 14 Hz), beta (14 - 30 Hz), and gamma (30 - 220 Hz) [74, 75]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to attention and search tasks of computation. Theta rhythms have been shown to modulate attention during auditory tasks [76, 77], visual search tasks [78–80], and in analysing the general attentional mechanism of the brain [17, 74, 75]. Intervals between theta peaks are finding stronger interpretation in the neuroscience literature as discretely sampling the environment for integration into the internal model of the brain. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [17, 76], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [78, 79]. Accordingly, the firing patterns which define the lowest level of information carried in the brain are regularly found to be gamma in scale ($\sim 30 - 100$ Hz) [81–85].

1.3.2 Gaze Trajectories

Healthy human gaze trajectories consist of different dynamical regimes. The most studied property of the pattern of gaze trajectories are their division into periods of fixation, interspersed by near-instantaneous jumps between fixations, these jumps being known as "saccades" [88–91]. Importantly, this behaviour of local movements interspersed with large discontinuous jumps suggest modelling eye movements through a Lévy flight. This method has a limited history in the literature, and fitting data to a simple phenomenological Lévy flight parametrised by the stable parameter α found limited use as a biomarker for ADHD [92]. Saccade metrics in general have been found to be valuable as non-invasive biomarkers for broader neural disorders, including schizophrenia, autism, ADHD, OCD, bipolar disorder, Parkinson's disease, and multiple sclerosis [93–97]. Periods of fixation have received less scholarly attention as a biomarker of cognition. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [98–101]. Examinations and literature reviews of these behaviours have often not considered their computational role in active sensing and efficient search.

Bibliography

(1) B. Wandell, S. Dumoulin and A. Brewer, in *Encyclopedia of Neuroscience*, Elsevier, 2009, p. 251.

(2) T. J. Andrews, S. D. Halpern and D. Purves, "Correlated Size Variations in Human Visual Cortex, Lateral Geniculate Nucleus, and Optic Tract", *The Journal of Neuroscience*, 1997, **17**, 2859–2868.

(3) S. Bhandari, P. Lencastre, R. Mathema, A. Szorkovszky, A. Yazidi and P. G. Lind, "Modeling eye gaze velocity trajectories using GANs with spectral loss for enhanced fidelity", *Scientific Reports*, 2025, **15**, 19929.

(4) S. Wu, X. Zhang, J. Chen, X. Zhang, Y. Xing and J. Feng, "GiTNet: A graph-based trajectory-informed network for gaze-supervised medical image segmentation", *Medical Image Analysis*, 2026, **112**, 104112.

(5) A. A. Abdelrahman, B. Al-Tawil and A. Al-Hamadi, "Deep Learning-Based Gaze Estimation: A Review", *Robotics*, 2026, **15**, 69.

(6) G. Bremer, N. Stein and M. Lappe, 2021 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR), IEEE, Taichung, Taiwan, 2021, pp. 19–28.

(7) S. Özdel, Y. Rong, B. M. Albaba, Y.-L. Kuo, X. Wang and E. Kasneci, Proceedings of the 2024 Symposium on Eye Tracking Research and Applications, ACM, Glasgow United Kingdom, 2024, pp. 1–9.

(8) A. A. Akinyelu and P. Blignaut, "Convolutional Neural Network-Based Technique for Gaze Estimation on Mobile Devices", *Frontiers in Artificial Intelligence*, 2022, **4**, 796825.

(9) M. Kümmerer, H. S. Khanuja and M. Bethge, *Modeling Saliency Dataset Bias*, en, arXiv:2505.10169 [cs.CV], 2025.

(10) M. Cornia, L. Baraldi, G. Serra and R. Cucchiara, 2016 23rd International Conference on Pattern Recognition (ICPR), IEEE, Cancun, 2016, pp. 3488–3493.

(11) M. Cornia, L. Baraldi, G. Serra and R. Cucchiara, "Predicting Human Eye Fixations via an LSTM-Based Saliency Attentive Model", *IEEE Transactions on Image Processing*, 2018, **27**, 5142–5154.

(12) M. Kämmerer and M. Bethge, "DeepGaze II: A Big Step Towards Explaining All Information in Image-Based Saliency", *Journal of Vision*, 2016, **16**, 330.

(13) X. Huang, C. Shen, X. Boix and Q. Zhao, 2015 IEEE International Conference on Computer Vision (ICCV), IEEE, Santiago, Chile, 2015, pp. 262–270.

(14) G. Bujia, M. Sclar, S. Vita, G. Solovey and J. E. Kamienkowski, "Modeling Human Visual Search in Natural Scenes: A Combined Bayesian Searcher and Saliency Map Approach", *Frontiers in Systems Neuroscience*, 2022, **16**, 882315.

(15) D. C. Knill and A. Pouget, "The Bayesian brain: the role of uncertainty in neural coding and computation", *Trends in Neurosciences*, 2004, **27**, 712–719.

(16) L. D. Costa, *Probabilistic Principles for Biophysics and Neuroscience: Entropy Production, Bayesian Mechanics & the Free-Energy Principle*, en, arXiv:2410.11735 [math-ph], 2024.

(17) E. Tan, S. V. Troller-Renfree, S. Morales, G. A. Buzzell, M. McSweeney, M. Antúnez and N. A. Fox, "Theta activity and cognitive functioning: Integrating evidence from resting-state and task-related developmental electroencephalography (EEG) research", *Developmental Cognitive Neuroscience*, 2024, **67**, 101404.

(18) G. Orbán, P. Berkes, J. Fiser and M. Lengyel, "Neural Variability and Sampling-Based Probabilistic Representations in the Visual Cortex", *Neuron*, 2016, **92**, 530–543.

(19) P. Hoyer and A. Hyvärinen, Advances in Neural Information Processing Systems, ed. S. Becker, S. Thrun and K. Obermayer, MIT Press, 2002, vol. 15.

(20) J. M. Beck, W. J. Ma, R. Kiani, T. Hanks, A. K. Churchland, J. Roitman, M. N. Shadlen, P. E. Latham and A. Pouget, "Probabilistic Population Codes for Bayesian Decision Making", *Neuron*, 2008, **60**, 1142–1152.

(21) J.-Q. Zhu, J. Sundh, J. Spicer, N. Chater and A. N. Sanborn, "The autocorrelated Bayesian sampler: A rational process for probability judgments, estimates, confidence intervals, choices, confidence judgments, and response times.", *Psychological Review*, 2024, **131**, 456–493.

(22) M. Mangalam, "The myth of the Bayesian brain", *European Journal of Applied Physiology*, 2025, **125**, 2643–2677.

(23) R. D. Lange, S. Shivkumar, A. Chattoraj and R. M. Haefner, "Bayesian encoding and decoding as distinct perspectives on neural coding", *Nat. Neurosci.*, 2023, **26**, 2063–2072.

(24) K. Friston, "The free-energy principle: a unified brain theory?", *Nat. Rev. Neurosci.*, 2010, **11**, 127–138.

(25) T. Parr, G. Pezzulo and K. Friston, *Active inference*, MIT Press, London, England, 2025.

(26) K. P. Körding and D. M. Wolpert, "Bayesian integration in sensorimotor learning", *Nature*, 2004, **427**, 244–247.

(27) R. P. N. Rao and D. H. Ballard, "Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects", *Nature Neuroscience*, 1999, **2**, 79–87.

(28) C. Koch and T. Poggio, "Predicting the visual world: silence is golden", *Nature Neuroscience*, 1999, **2**, 9–10.

(29) U. Hassler, *Stochastic Processes and Calculus: An Elementary Introduction with Applications*, Springer International Publishing, Cham, 2016.

(30) S. Chandrasekhar, "Dynamical Friction. I. General Considerations: The Coefficient of Dynamical Friction", *The Astrophysical Journal*, 1943, **97**, 255–262.

(31) D. Merritt, P. Berczik and F. Laun, "Brownian Motion of Black Holes in Dense Nuclei", *The Astronomical Journal*, 2007, **133**, 553–563.

(32) A. Cunchala, *A Basic Overview of Various Stochastic Approaches to Financial Modeling With Examples*, 2024.

(33) S. Brevitt, A. Schulz, D. Pegler, H. Kantz and R. Klages, "Singularity of Lévy walks in the lifted Pomeau–Manneville map", *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 2025, **35**, 013151.

(34) R. Klages, in *From Hamiltonian Chaos to Complex Systems: A Nonlinear Physics Approach*, ed. X. Leoncini and M. Leonetti, Springer New York, New York, NY, 2013, pp. 3–42.

(35) J. Klafter and I. M. Sokolov, "Anomalous diffusion spreads its wings", *Physics World*, 2005, **18**, 29–32.

(36) J. W. Kirchner, X. Feng and C. Neal, "Fractal stream chemistry and its implications for contaminant transport in catchments", *Nature*, 2000, **403**, 524–527.

(37) S. C. Kou and X. S. Xie, "Generalized Langevin equation with fractional Gaussian noise: subdiffusion within a single protein molecule", *Phys. Rev. Lett.*, 2004, **93**, 180603.

(38) G. M. Viswanathan, S. V. Buldyrev, S. Havlin, M. G. E. Da Luz, E. P. Raposo and H. E. Stanley, "Optimizing the success of random searches", *Nature*, 1999, **401**, 911–914.

(39) N. E. Humphries and D. W. Sims, "Optimal foraging strategies: Lévy walks balance searching and patch exploitation under a very broad range of conditions", *Journal of Theoretical Biology*, 2014, **358**, 179–193.

(40) W. Campeau, A. M. Simons and B. Stevens, "The evolutionary maintenance of Lévy flight foraging", *PLOS Computational Biology*, 2022, **18**, ed. M. G. E. Da Luz, e1009490.

(41) D. W. Sims, N. E. Humphries, N. Hu, V. Medan and J. Berni, "Optimal searching behaviour generated intrinsically by the central pattern generator for locomotion", *eLife*, 2019, **8**, e50316.

(42) A. Clementi, F. d'Amore, G. Giakkoupis and E. Natale, Proceedings of the 2021 ACM Symposium on Principles of Distributed Computing, ACM, Virtual Event Italy, 2021, pp. 81–91.

(43) A. M. Reynolds and M. A. Frye, "Free-Flight Odor Tracking in Drosophila Is Consistent with an Optimal Intermittent Scale-Free Search", *PLoS ONE*, 2007, **2**, ed. S. Rands, e354.

(44) S. Focardi and J. G. Cecere, "The Lévy flight foraging hypothesis in a pelagic seabird", *Journal of Animal Ecology*, 2014, **83**, ed. G. Hays, 353–364.

(45) N. E. Humphries, H. Weimerskirch and D. W. Sims, "A new approach for objective identification of turns and steps in organism movement data relevant to random walk modelling", *Methods in Ecology and Evolution*, 2013, **4**, ed. R. Freckleton, 930–938.

(46) D. W. Sims, E. J. Southall, N. E. Humphries, G. C. Hays, C. J. A. Bradshaw, J. W. Pitchford, A. James, M. Z. Ahmed, A. S. Brierley, M. A. Hindell, D. Morritt, M. K. Musyl, D. Righton, E. L. C. Shepard, V. J. Wearmouth, R. P. Wilson, M. J. Witt and J. D. Metcalfe, "Scaling laws of marine predator search behaviour", *Nature*, 2008, **451**, 1098–1102.

(47) N. E. Humphries, N. Queiroz, J. R. M. Dyer, N. G. Pade, M. K. Musyl, K. M. Schaefer, D. W. Fuller, J. M. Brunnschweiler, T. K. Doyle, J. D. R. Houghton, G. C. Hays, C. S. Jones, L. R. Noble, V. J. Wearmouth, E. J. Southall and D. W. Sims, "Environmental context explains Lévy and Brownian movement patterns of marine predators", *Nature*, 2010, **465**, 1066–1069.

(48) A. O. Gautestad and A. Mysterud, "The Lévy flight foraging hypothesis: forgetting about memory may lead to false verification of Brownian motion", *Movement Ecology*, 2013, **1**, 9.

(49) G. H. Pyke, A. G. Kepert and W. J. Chivers, "Lévy foraging hypothesis is a model artefact", *Ecological Modelling*, 2026, **519**, 111647.

(50) G. H. Pyke, "Understanding movements of organisms: it's time to abandon the Lévy foraging hypothesis", *Methods in Ecology and Evolution*, 2015, **6**, ed. L. Giuggioli, 1–16.

(51) S. Benhamou and J. Collet, "Ultimate failure of the Lévy Foraging Hypothesis: Two-scale searching strategies outperform scale-free ones even when prey are scarce and cryptic", *Journal of Theoretical Biology*, 2015, **387**, 221–227.

(52) A. Reynolds, "Liberating Lévy walk research from the shackles of optimal foraging", *Physics of Life Reviews*, 2015, **14**, 59–83.

(53) M. van den Bosch, "Notes on stochastic integration theory with respect to càdlàg semimartingales and a brief introduction to Lévy processes", 2024, 1–70.

(54) P. Soulier, "The tail process and tail measure of continuous time regularly varying stochastic processes", *Extremes*, 2022, **25**, 107–173.

(55) V. S. Borkar, in *Probability Theory: An Advanced Course*, Springer New York, New York, NY, 1995, pp. 109–129.

(56) A. E. Kyprianou, in *Fluctuations of Lévy Processes with Applications: Introductory Lectures*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2014, pp. 33–36.

(57) J. Delacour, J. Lévy and D. Mercier, "A 'Neural Sampling Theory (NST)' of learning and memory mechanisms", *Biosystems*, 1997, **44**, 231–244.

(58) K. Cranmer, G. Kanwar, S. Racanière, D. J. Rezende and P. E. Shanahan, "Advances in machine-learning-based sampling motivated by lattice quantum chromodynamics", *Nature Reviews Physics*, 2023, **5**, 526–535.

(59) J. Bohannon, "A computer program just ranked the most influential brain scientists of the modern era", *Science*, 2016.

(60) K. Friston, "Does predictive coding have a future?", *Nature Neuroscience*, 2018, **21**, 1019–1021.

(61) R. D. Lange, S. Shivkumar, A. Chattoraj and R. M. Haefner, "Bayesian Encoding and Decoding as Distinct Perspectives on Neural Coding", 2020.

(62) Y. Terada and T. Toyoizumi, "Chaotic neural dynamics facilitate probabilistic computations through sampling", *Proceedings of the National Academy of Sciences*, 2024, **121**, e2312992121.

(63) X. Dong and S. Wu, "Neural Sampling in Hierarchical Exponential-family Energy-based Models".

(64) V. Koren, S. Blanco Malerba, T. Schwalger and S. Panzeri, "Efficient coding in biophysically realistic excitatory-inhibitory spiking networks", *eLife*, 2025, **13**, RP99545.

(65) J. A. Kilgore, J. D. Kopsick, G. A. Ascoli and G. C. Adam, "Biologically-informed excitatory and inhibitory ratio for robust spiking neural network training", *Scientific Reports*, 2025, **15**, 24798.

(66) H. F. Song, G. R. Yang and X.-J. Wang, "Training Excitatory-Inhibitory Recurrent Neural Networks for Cognitive Tasks: A Simple and Flexible Framework", *PLOS Computational Biology*, 2016, **12**, ed. O. Sporns, e1004792.

(67) V. Tarasov, "General Fractional Dynamics", *Mathematics*, 2021, **9**, 1464.

(68) R. M. Neal, "Sampling from multimodal distributions using tempered transitions", *Statistics and Computing*, 1996, **6**, 353–366.

(69) K. Łatuszyński, M. T. Moores and T. Stumpf-Fétizon, *MCMC for multi-modal distributions*, en, arXiv:2501.05908 [stat], 2025.

(70) U. Simsekli, "Fractional Langevin Monte Carlo: Exploring Lévy Driven Stochastic Differential Equations for Markov Chain Monte Carlo".

(71) R. Metzler and J. Klafter, "The random walk's guide to anomalous diffusion: a fractional dynamics approach", *Physics Reports*, 2000, **339**, 1–77.

(72) N. Ye and Z. Zhu, Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18, International Joint Conferences on Artificial Intelligence Organization, 2018, pp. 3019–3025.

(73) Y. Qi and P. Gong, "Fractional neural sampling as a theory of spatiotemporal probabilistic computations in neural circuits", *Nature Communications*, 2022, **13**, 4572.

(74) J. Fan, J. Byrne, M. S. Worden, K. G. Guise, B. D. McCandliss, J. Fossella and M. I. Posner, "The Relation of Brain Oscillations to Attentional Networks", *Journal of Neuroscience*, 2007, **27**, 6197–6206.

(75) W. Klimesch, "The frequency architecture of brain and brain body oscillations: an analysis", *European Journal of Neuroscience*, 2018, **48**, 2431–2453.

(76) W. Wang, G. Liu and Y. Xi, Proceedings of the 6th International Conference on High Performance Compilation, Computing and Communications, ACM, Jilin China, 2022, pp. 163–167.

(77) X. Yang, C. Ying, L. Zhu and W. Wenjing, "The neural oscillations in delta- and theta-bands contribute to divided attention in audiovisual integration", *Perception*, 2024, **53**, 44–60.

(78) T. Kawashima, M. J. Hayashi and K. Amano, *Attentional rhythmic blink: Theta/Alpha balance in neural oscillations determines the rhythmicity in visual sampling*, en, 2022.

(79) G. Spyropoulos, C. A. Bosman and P. Fries, "A theta rhythm in macaque visual cortex and its attentional modulation", *Proceedings of the National Academy of Sciences*, 2018, **115**, DOI: 10.1073/pnas.1719433115.

(80) C. Mo, J. Lu, W. B., J. J., L. H. and F. F., "Competing rhythmic neural representations of orientations during concurrent attention to multiple orientation features", 2019.

(81) R. T. Canolty, E. Edwards, S. S. Dalal, M. Soltani, S. S. Nagarajan, H. E. Kirsch, M. S. Berger, N. M. Barbaro and R. T. Knight, "High Gamma Power Is Phase-Locked to Theta Oscillations in Human Neocortex", *Science*, 2006, **313**, 1626–1628.

(82) B. Harris and P. Gong, "Nested spatiotemporal theta-gamma waves organize hierarchical processing across the mouse visual cortex", *Nat. Commun.*, 2026, **17**.

(83) M. Ursino and G. Pirazzini, "Theta–gamma coupling as a ubiquitous brain mechanism: implications for memory, attention, dreaming, imagination, and consciousness", *Current Opinion in Behavioral Sciences*, 2024, **59**, 101433.

(84) H. Brooks, M. S. Goodman, C. R. Bowie, R. Zomorrodi, D. M. Blumberger, M. A. Butters, Z. J. Daskalakis, C. E. Fischer, A. Flint, N. Herrmann, S. Kumar, L. Mah, B. H. Mulsant, B. G. Pollock, A. N. Voineskos, T. K. Rajji, on behalf of the PACt-MD Study Group, B. H. Mulsant, T. K. Rajji, N. Herrmann, B. G. Pollock, L. Lourenco, D. M. Blumberger, C. R. Bowie, M. A. Butters, C. E. Fischer, A. Flint, D. Gallagher, A. Golas, A. Graff, J. L. Kennedy, S. Kumar, L. Mah, S. Ovaysikia, M. Rapoport, K. E. Thorpe, N. P. L. G. Verhoeff and A. N. Voineskos, "Theta–gamma coupling and ordering information: a stable brain–behavior relationship across cognitive tasks and clinical conditions", *Neuropsychopharmacology*, 2020, **45**, 2038–2047.

(85) A. B. L. Tort, R. W. Komorowski, J. R. Manns, N. J. Kopell and H. Eichenbaum, "Theta–gamma coupling increases during the learning of item–context associations", *Proceedings of the National Academy of Sciences*, 2009, **106**, 20942–20947.

(86) J. E. Lisman and O. Jensen, "The Theta-Gamma Neural Code", *Neuron*, 2013, **77**, 1002–1016.

(87) D. McLelland and R. VanRullen, "Theta-Gamma Coding Meets Communication-through-Coherence: Neuronal Oscillatory Multiplexing Theories Reconciled", *PLOS Computational Biology*, 2016, **12**, ed. B. S. Gutkin, e1005162.

(88) C. J. Erkelens, "Coordination of smooth pursuit and saccades", *Vision Research*, 2006, **46**, 163–170.

(89) H. S. Seung, "How the brain keeps the eyes still", *Proceedings of the National Academy of Sciences*, 1996, **93**, 13339–13344.

(90) P. Lencastre, Y. S. Bystryk, A. Yazidi, S. Denisov and P. G. Lind, "Dynamical law behind eye movements: Distinguishing between Lévy and intermittent strategies", *Physical Review Research*, 2025, **7**, 043042.

(91) J. T. Benitezy, "Eye-tracking and optokinetic tests: Diagnostic significance in peripheral and central vestibular disorders", *The Laryngoscope*, 1970, **80**, 834–848.

(92) C. Papanikolaou, A. Sharma, P. G. Lind and P. Lencastre, "Lévy Flight Model of Gaze Trajectories to Assist in ADHD Diagnoses", *Entropy*, 2024, **26**, 392.

(93) K. Morita, K. Miura, M. Fujimoto, H. Yamamori, Y. Yasuda, M. Iwase, K. Kasai and R. Hashimoto, "Eye movement as a biomarker of schizophrenia: Using an integrated eye movement score", *Psychiatry and Clinical Neurosciences*, 2017, **71**, 104–114.

(94) J. Bittencourt, Velasques, Teixeira, Basile, Salles, Nardi, Budde, Cagy, Piedade and Ribeiro, "Saccadic eye movement applications for psychiatric disorders", *Neuropsychiatric Disease and Treatment*, 2013, 1393.

(95) J. Zhu, L. Zhou, Y. Zhou, Y. Lin, Y. Cai, J. Wu and C. Shi, "Diagnosis of schizophrenia by integrated saccade scores and associations with psychiatric symptoms, and functioning", *Medicine*, 2024, **103**, e39935.

(96) S. Caldani, M. P. Bucci, J.-C. Lamy, M. Seassau, N. Bendjemaa, R. Gadel, R. Gaillard, M.-O. Krebs and I. Amado, “Saccadic eye movements as markers of schizophrenia spectrum: Exploration in at-risk mental states”, *Schizophrenia Research*, 2017, **181**, 30–37.

(97) Y. L. Wang, M. Kapoor, J. Fielding, S. Reddel, C. Zhu, M. Clough, M. Botrous, M. Monif and A. Van Der Walt, “Saccadic eye movements in neurological disease: cognitive mechanisms and clinical applications”, *Journal of Neurology*, 2025, **272**, 539.

(98) F. Van Ede, “Looking into working memory through micro eye movements”, *Trends in Cognitive Sciences*, 2026, S1364661326001282.

(99) L. Graham, J. Das, R. Vitorio, C. McDonald, R. Walker, A. Godfrey, R. Morris and S. Stuart, “Ocular microtremor: a structured review”, *Experimental Brain Research*, 2023, **241**, 2191–2203.

(100) R. Engbert, K. Mergenthaler, P. Sinn and A. Pikovsky, “An integrated model of fixational eye movements and microsaccades”, *Proceedings of the National Academy of Sciences*, 2011, **108**, DOI: 10.1073/pnas.1102730108.

(101) A. Spauschus, J. Marsden, D. M. Halliday, J. R. Rosenberg and P. Brown, “The origin of ocular microtremor in man”, *Experimental Brain Research*, 1999, **126**, 556–562.